

Airline network analysis

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AY network

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Challenges

abc

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Network optimization.

NRPS does this.

$$Rev = \langle f, x \rangle \rightarrow \max,$$

$$Ax \leq c,$$

$$0 \leq x \leq \mu.$$

where

- ▶ μ is vector of ODIFs demands.
- ▶ A is product matrix.
- ▶ c is vector of leg capacities.
- ▶ f is vector of ODIF fares.
- ▶ x is vector of allocations.

Controlled vs Uncontrolled

Controlled:

- ▶ f is our price (BMs).
- ▶ A is about where we fly and SigOD list.
- ▶ c is our capacity on leg.

Uncontrolled:

- ▶ μ is demand on ODIFs.